

ZERO-TO-HERO FORMULA NOTES

Differentiation & Integration Formula Master Notes

A clean, beginner-friendly formula guide for Ordinary Differential Equations. Designed for easy revision, clear examples, and fast method recognition.

Differentiation

Rules, common derivatives, chain rule, product rule, implicit differentiation.

Integration

Power rule, logs, trig, substitution, parts, partial fractions basics.

ODE Ready

Separable equations, integrating factor, homogeneous and exact formulas.

Beginner Friendly

Examples included with clean alignment and readable spacing.

Study tip: Learn the formulas in the priority order at the end of this PDF. Do not memorize everything in one day.

Differentiation & Integration Formula Master Notes

| Zero-to-Hero Formula Sheet for Differential Equations

Purpose: This file contains the main differentiation and integration formulas you must remember before studying Ordinary Differential Equations (ODE).

Level: Beginner-friendly.

Assumption: You know basic algebra, differentiation, and integration, but you need one clean formula guide.

How to Use This File

Do not try to memorize everything in one day.

Study in this order:

1. Basic algebra rules
 2. Differentiation rules
 3. Common derivatives
 4. Integration rules
 5. Common integrals
 6. Trigonometric identities
 7. Exponential and logarithm rules
 8. Differential equation related formulas
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1. Basic Algebra Rules You Must Remember

1.1 Power rules

$$x^a \cdot x^b = x^{(a+b)}$$

Example:

$$x^2 \cdot x^3 = x^5$$

$$x^a / x^b = x^{(a-b)}$$

Example:

$$x^5 / x^2 = x^3$$

$$(x^a)^b = x^{(ab)}$$

Example:

$$(x^2)^3 = x^6$$

$$x^0 = 1$$

$$x^{-a} = 1/x^a$$

Example:

$$x^{-2} = 1/x^2$$

$$\sqrt{x} = x^{(1/2)}$$

$$1/\sqrt{x} = x^{(-1/2)}$$

1.2 Fraction rules

$$a/b + c/b = (a+c)/b$$

$$a/b - c/b = (a-c)/b$$

$$(a/b)(c/d) = ac/bd$$

$$(a/b) \div (c/d) = (a/b)(d/c)$$

1.3 Completing the square

This is useful for integrals such as:

$$\int dx/(x^2 + x + 1)$$

Formula:

$$x^2 + bx + c = (x + b/2)^2 + c - b^2/4$$

Example:

$$x^2 + x + 1$$

$$= (x + 1/2)^2 + 1 - 1/4$$

$$= (x + 1/2)^2 + 3/4$$

2. Differentiation Basics

2.1 Meaning of derivative

The derivative of y with respect to x is written as:

$$dy/dx$$

It means the rate of change of y when x changes.

If:

$$y = f(x)$$

then:

$$dy/dx = f'(x)$$

3. Main Differentiation Rules

3.1 Constant rule

If:

$$y = C$$

where C is a constant, then:

$$dy/dx = 0$$

Example:

$$d/dx(7) = 0$$

3.2 Power rule

$$d/dx(x^n) = nx^{(n-1)}$$

Example:

$$d/dx(x^3) = 3x^2$$

$$d/dx(x^{-2}) = -2x^{-3}$$

$$d/dx(\sqrt{x}) = d/dx(x^{(1/2)}) = (1/2)x^{(-1/2)}$$

So:

$$d/dx(\sqrt{x}) = 1/(2\sqrt{x})$$

3.3 Constant multiple rule

$$d/dx[cf(x)] = c f'(x)$$

Example:

$$d/dx(5x^3) = 5(3x^2) = 15x^2$$

3.4 Sum rule

$$d/dx[f(x) + g(x)] = f'(x) + g'(x)$$

Example:

$$d/dx(x^2 + \sin x) = 2x + \cos x$$

3.5 Difference rule

$$d/dx[f(x) - g(x)] = f'(x) - g'(x)$$

Example:

$$d/dx(x^3 - e^x) = 3x^2 - e^x$$

3.6 Product rule

Use this when two functions are multiplied.

$$d/dx[uv] = u(dv/dx) + v(du/dx)$$

Short form:

$$(uv)' = uv' + vu'$$

Example:

$$y = x^2 \sin x$$

Let:

$$u = x^2, \quad v = \sin x$$

Then:

$$u' = 2x, \quad v' = \cos x$$

So:

$$y' = x^2 \cos x + \sin x(2x)$$

Final:

$$y' = x^2 \cos x + 2x \sin x$$

3.7 Quotient rule

Use this when one function is divided by another.

$$d/dx(u/v) = [v(du/dx) - u(dv/dx)] / v^2$$

Short form:

$$(u/v)' = (vu' - uv')/v^2$$

Example:

$$y = x^2/\sin x$$

Let:

$$u = x^2, \quad v = \sin x$$

Then:

$$u' = 2x, \quad v' = \cos x$$

So:

$$y' = [\sin x(2x) - x^2 \cos x]/\sin^2 x$$

3.8 Chain rule

Use this when there is a function inside another function.

$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$$

Simple memory:

Differentiate outside, keep inside same, then multiply by derivative of inside.

Example:

$$y = (3x + 2)^5$$

Outside function is power 5. Inside is $3x+2$.

$$y' = 5(3x+2)^4 \cdot 3$$

Final:

$$y' = 15(3x+2)^4$$

Another example:

$$y = \sin(x^2)$$

$$y' = \cos(x^2) \cdot 2x$$

Final:

$$y' = 2x \cos(x^2)$$

3.9 Implicit differentiation

Use this when x and y are mixed together.

Example:

$$x^2 + y^2 = 25$$

Differentiate both sides with respect to x :

$$2x + 2y \, dy/dx = 0$$

Then:

$$2y \, dy/dx = -2x$$

$$dy/dx = -x/y$$

Important:

Whenever differentiating a term with y , multiply by dy/dx .

Examples:

$$d/dx(y) = dy/dx$$

$$d/dx(y^2) = 2y \, dy/dx$$

$$d/dx(\sin y) = \cos y \, dy/dx$$

$$d/dx(e^y) = e^y \, dy/dx$$

3.10 Logarithmic differentiation

Useful for complicated powers/products.

If:

$$y = x^x$$

Take $\ln$ on both sides:

$$\ln y = x \ln x$$

Differentiate:

$$(1/y)y' = \ln x + 1$$

So:

$$y' = y(\ln x + 1)$$

Since $y = x^x$:

$$y' = x^x(\ln x + 1)$$

4. Common Differentiation Formulas

4.1 Polynomial and power functions

Function	Derivative
C	0
x	1
x^2	$2x$
x^3	$3x^2$
x^n	$nx^{(n-1)}$
$1/x$	$-1/x^2$
$\sqrt{x}$	$1/(2\sqrt{x})$

4.2 Exponential functions

Function	Derivative
e^x	e^x
$e^{(ax)}$	$ae^{(ax)}$
a^x	$a^x \ln a$

Examples:

$$d/dx(e^{(3x)}) = 3e^{(3x)}$$

$$d/dx(2^x) = 2^x \ln 2$$

4.3 Logarithmic functions

Function	Derivative
$\ln x$	$1/x$
$\ln x$	x $1/x$
$\log_a x$	$1/(x \ln a)$
$\ln(f(x))$	$f'(x)/f(x)$

Examples:

$$d/dx[\ln(3x+1)] = 3/(3x+1)$$

$$d/dx[\ln(x^2+1)] = 2x/(x^2+1)$$

4.4 Trigonometric functions

Function	Derivative
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

Note: Some books write $\operatorname{cosec} x$; others write $\csc x$. They are the same.

$$\operatorname{cosec} x = \csc x$$

4.5 Trigonometric chain rule formulas

Function	Derivative
$\sin(ax+b)$	$a \cos(ax+b)$
$\cos(ax+b)$	$-a \sin(ax+b)$
$\tan(ax+b)$	$a \sec^2(ax+b)$
$\sec(ax+b)$	$a \sec(ax+b)\tan(ax+b)$

Examples:

$$d/dx[\sin(5x)] = 5\cos(5x)$$

$$d/dx[\cos(2x+1)] = -2\sin(2x+1)$$

4.6 Inverse trigonometric functions

Function	Derivative
$\sin^{-1}x$ or $\arcsin x$	$1/\sqrt{1-x^2}$
$\cos^{-1}x$ or $\arccos x$	$-1/\sqrt{1-x^2}$
$\tan^{-1}x$ or $\arctan x$	$1/(1+x^2)$
$\cot^{-1}x$	$-1/(1+x^2)$
$\sec^{-1}x$	$1/(\sqrt{x^2-1})$
$\operatorname{cosec}^{-1}x$	$-1/(\sqrt{x^2-1})$

Very important for ODE:

$$d/dx[\tan^{-1}x] = 1/(1+x^2)$$

5. Higher Derivatives

5.1 First derivative

$$y' = dy/dx$$

5.2 Second derivative

$$y'' = d^2y/dx^2$$

5.3 Third derivative

$$y''' = d^3y/dx^3$$

Example:

$$y = x^4$$

First derivative:

$$y' = 4x^3$$

Second derivative:

$$y'' = 12x^2$$

Third derivative:

$$y''' = 24x$$

Fourth derivative:

$$y'''' = 24$$

6. Integration Basics

6.1 Meaning of integration

Integration is the reverse of differentiation.

If:

$$d/dx[F(x)] = f(x)$$

then:

$$\int f(x) \, dx = F(x) + C$$

The $+ C$ is important because derivative of any constant is zero.

Example:

$$d/dx(x^2 + 5) = 2x$$

$$d/dx(x^2 - 100) = 2x$$

So:

$$\int 2x \, dx = x^2 + C$$

7. Main Integration Rules

7.1 Constant rule

$$\int a \, dx = ax + C$$

Example:

$$\int 5 \, dx = 5x + C$$

7.2 Power rule

For $n \neq -1$:

$$\int x^n \, dx = x^{(n+1)}/(n+1) + C$$

Examples:

$$\int x^2 \, dx = x^3/3 + C$$

$$\int x^5 \, dx = x^6/6 + C$$

$$\int \sqrt{x} \, dx = \int x^{(1/2)} \, dx = x^{(3/2)}/(3/2) + C$$

$$= (2/3)x^{(3/2)} + C$$

7.3 Special power rule when $n = -1$

$$\int 1/x \, dx = \ln|x| + C$$

Important:

$$\int x^{-1} \, dx = \ln|x| + C$$

Not:

$$x^0/0$$

because division by zero is not allowed.

7.4 Constant multiple rule

$$\int cf(x) \, dx = c \int f(x) \, dx$$

Example:

$$\int 7x^2 \, dx = 7 \int x^2 \, dx$$

$$= 7x^3/3 + C$$

7.5 Sum rule

$$\int [f(x) + g(x)] \, dx = \int f(x) \, dx + \int g(x) \, dx$$

Example:

$$\int (x^2 + \cos x) \, dx = x^3/3 + \sin x + C$$

7.6 Difference rule

$$\int [f(x) - g(x)] \, dx = \int f(x) \, dx - \int g(x) \, dx$$

Example:

$$\int (x^3 - e^x) \, dx = x^4/4 - e^x + C$$

8. Common Integration Formulas

8.1 Polynomial and power integrals

Function	Integral
1	$x + C$
x	$x^2/2 + C$
x^2	$x^3/3 + C$
x^3	$x^4/4 + C$
$x^n, n \neq -1$	$x^{(n+1)}/(n+1) + C$
$1/x$	$\ln x + C$
$1/x^2$	$-1/x + C$
$\sqrt{x}$	$(2/3)x^{(3/2)} + C$
$1/\sqrt{x}$	$2\sqrt{x} + C$

8.2 Exponential integrals

Function	Integral
e^x	$e^x + C$
$e^{(ax)}$	$e^{(ax)}/a + C, a \neq 0$
a^x	$a^x/\ln a + C$

Examples:

$\int e^{(3x)} dx = e^{(3x)}/3 + C$

$\int 2^x dx = 2^x/\ln 2 + C$

8.3 Logarithmic related integrals

Function	Integral		
$1/x$	$\ln$	x	$+ C$
$1/(ax+b)$	$(1/a)\ln$	$ax+b$	$+ C$
$f'(x)/f(x)$	$\ln$	$f(x)$	$+ C$

Examples:

$$\int 1/(3x+2) \, dx = (1/3)\ln|3x+2| + C$$

$$\int 2x/(x^2+1) \, dx = \ln(x^2+1) + C$$

8.4 Trigonometric integrals

Function	Integral
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\operatorname{cosec}^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$
$\operatorname{cosec} x \cot x$	$-\operatorname{cosec} x + C$

Examples:

$$\int \cos x \, dx = \sin x + C$$

$$\int \sin x \, dx = -\cos x + C$$

8.5 Trigonometric integrals with $ax+b$

Function	Integral
$\sin(ax+b)$	$-\cos(ax+b)/a + C$
$\cos(ax+b)$	$\sin(ax+b)/a + C$
$\sec^2(ax+b)$	$\tan(ax+b)/a + C$
$\operatorname{cosec}^2(ax+b)$	$-\cot(ax+b)/a + C$

Examples:

$$\int \sin(3x) \, dx = -\cos(3x)/3 + C$$

$$\int \cos(5x+1) \, dx = \sin(5x+1)/5 + C$$

8.6 Important inverse trigonometric integrals

Integral	Answer
$\int dx/(1+x^2)$	$\tan^{-1}x + C$
$\int dx/\sqrt{1-x^2}$	$\sin^{-1}x + C$
$\int dx/(a^2+x^2)$	$(1/a)\tan^{-1}(x/a) + C$
$\int dx/\sqrt{a^2-x^2}$	$\sin^{-1}(x/a) + C$

Very important:

$$\int dx/(1+x^2) = \tan^{-1}x + C$$

Example:

$$\int dx/(4+x^2)$$

Here:

$$a^2 = 4, \text{ so } a = 2$$

Therefore:

$\int dx/(4+x^2) = (1/2)\tan^{-1}(x/2) + C$

8.7 Integrals involving square roots

Integral	Answer		
$\int dx/\sqrt{a^2-x^2}$	$\sin^{-1}(x/a) + C$		
$\int dx/\sqrt{x^2+a^2}$	$\ln$	$x + \sqrt{x^2+a^2}$	$+ C$
$\int dx/\sqrt{x^2-a^2}$	$\ln$	$x + \sqrt{x^2-a^2}$	$+ C$

9. Integration by Substitution

9.1 When to use

Use substitution when the integral has a function and its derivative.

Pattern:

$$\int f'(x)[f(x)]^n dx$$

or:

$$\int f'(x)/f(x) dx$$

9.2 Steps

1. Let $u = \text{inside function}$.
2. Find du/dx .
3. Rewrite dx part using du .
4. Integrate in terms of u .
5. Replace u with original expression.

9.3 Example 1

Solve:

$$\int 2x(x^2+1)^5 dx$$

Let:

$$u = x^2 + 1$$

Then:

$$du/dx = 2x$$

$$du = 2x dx$$

So:

$$\int 2x(x^2+1)^5 dx = \int u^5 du$$

$$= u^6/6 + C$$

Replace u :

$$= (x^2+1)^6/6 + C$$

9.4 Example 2

Solve:

$$\int 3/(3x+2) dx$$

Let:

$$u = 3x + 2$$

Then:

$$du = 3 dx$$

So:

$$\int 3/(3x+2) dx = \int du/u$$

$$= \ln|u| + C$$

Final:

$$\ln|3x+2| + C$$

10. Integration by Parts

10.1 Formula

$$\int u \, dv = uv - \int v \, du$$

Use this when the integral is a product, such as:

$$\int x e^x \, dx$$

$$\int x \sin x \, dx$$

$$\int \ln x \, dx$$

10.2 Choosing u

Use LIATE rule:

1. Logarithmic: $\ln x$
2. Inverse trig: $\tan^{-1}x$
3. Algebraic: x , x^2
4. Trig: $\sin x$, $\cos x$
5. Exponential: e^x

Choose the earlier one as u .

10.3 Example

Solve:

$$\int x e^x \, dx$$

Choose:

$$u = x$$

$$dv = e^x dx$$

Then:

$$du = dx$$

$$v = e^x$$

Apply formula:

$$\int x e^x dx = x e^x - \int e^x dx$$

$$= x e^x - e^x + C$$

Final:

$$e^x(x-1) + C$$

11. Partial Fractions Basics

Partial fractions are used for integrals like:

$$\int dx / [(x+1)(x+2)]$$

11.1 Basic form

If:

$$1 / [(x+a)(x+b)]$$

write:

$$1 / [(x+a)(x+b)] = A / (x+a) + B / (x+b)$$

Then solve for **A** and **B**.

11.2 Example

Solve:

$$\int dx / [(x+1)(x+2)]$$

Write:

$$1 / [(x+1)(x+2)] = A / (x+1) + B / (x+2)$$

Multiply by $(x+1)(x+2)$:

$$1 = A(x+2) + B(x+1)$$

Put $x = -2$:

$$1 = A(0) + B(-1)$$

$$B = -1$$

Put $x = -1$:

$$1 = A(1) + B(0)$$

$$A = 1$$

So:

$$\int dx/[(x+1)(x+2)] = \int [1/(x+1) - 1/(x+2)]dx$$

$$= \ln|x+1| - \ln|x+2| + C$$

Final:

$$\ln|(x+1)/(x+2)| + C$$

12. Special Integrals Used in ODE

These appear often in differential equations.

12.1 Basic ODE integrals

Integral	Answer
$\int P(x) \, dx$	Used in integrating factor
$\int dy/y$	$\ln y + C$
$\int dx/x$	$\ln x + C$
$\int y^n \, dy$	$y^{(n+1)/(n+1)} + C, \quad n \neq -1$
$\int x^n \, dx$	$x^{(n+1)/(n+1)} + C, \quad n \neq -1$

12.2 Exponential conversion from logarithm

Very important for separable equations.

If:

$$\ln|y| = F(x) + C$$

then:

$$|y| = e^{(F(x)+C)}$$

Using:

$$e^{(a+b)} = e^a e^b$$

we get:

$$|y| = e^C e^{(F(x))}$$

Since e^C is just another constant, write:

$$y = Ce^{(F(x))}$$

Example:

$$\ln|y| = x^2/2 + C$$

Then:

$$y = Ce^{(x^2/2)}$$

12.3 Integrating factor formula

For:

$$dy/dx + P(x)y = Q(x)$$

Integrating factor:

$$I.F. = e^{(\int P(x)dx)}$$

Then:

$$y(I.F.) = \int Q(x)(I.F.)dx + C$$

Common $P(x)$ cases:

$P(x)$	$\int P \, dx$	I.F.	
a	ax	$e^{(ax)}$	
$1/x$	$\ln$	x	x
$2/x$	$2\ln$	x	x^2
$-1/x$	$-\ln$	x	$1/x$
$-3/x$	$-3\ln$	x	x^{-3}

Important simplification:

$$e^{(\ln x)} = x$$

$$e^{(2\ln x)} = x^2$$

$$e^{(-\ln x)} = 1/x$$

13. Trigonometric Identities to Remember

13.1 Basic identities

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \operatorname{cosec}^2 x$$

13.2 Quotient identities

$$\tan x = \sin x / \cos x$$

$$\cot x = \cos x / \sin x$$

13.3 Reciprocal identities

$$\sec x = 1/\cos x$$

$$\operatorname{cosec} x = 1/\sin x$$

$$\cot x = 1/\tan x$$

13.4 Double angle identities

$$\sin 2x = 2\sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x$$

Also:

$$\cos 2x = 1 - 2\sin^2x$$

$$\cos 2x = 2\cos^2x - 1$$

13.5 Useful rearrangements

From:

$$\sin^2x + \cos^2x = 1$$

we get:

$$\sin^2x = 1 - \cos^2x$$

$$\cos^2x = 1 - \sin^2x$$

From:

$$1 + \tan^2x = \sec^2x$$

we get:

$$\tan^2x = \sec^2x - 1$$

14. Exponential and Logarithm Rules

14.1 Exponential rules

$$e^a \cdot e^b = e^{(a+b)}$$

$$e^a / e^b = e^{(a-b)}$$

$$(e^a)^b = e^{(ab)}$$

$$e^0 = 1$$

$$e^{(\ln x)} = x$$

$$\ln(e^x) = x$$

14.2 Logarithm rules

$$\ln(ab) = \ln a + \ln b$$

$$\ln(a/b) = \ln a - \ln b$$

$$\ln(a^n) = n \ln a$$

$$\ln 1 = 0$$

$$\ln e = 1$$

14.3 Important ODE constant idea

If C is arbitrary, then these are also arbitrary constants:

$$C + 5$$

$$2C$$

$$-C$$

$$e^C$$

So we often rename constants.

Example:

$$\ln|y| = x + C$$

Exponentiate:

$$|y| = e^{(x+C)}$$

$$|y| = e^x e^C$$

Since e^C is just a constant:

$$y = Ce^x$$

15. Derivatives Needed for Differential Equations

These are especially useful when checking solutions.

15.1 Exponential solution checking

If:

$$y = e^{(\lambda x)}$$

then:

$$y' = \lambda e^{(\lambda x)}$$

$$y'' = \lambda^2 e^{(\lambda x)}$$

$$y''' = \lambda^3 e^{(\lambda x)}$$

In general:

$$y^{(n)} = \lambda^n e^{(\lambda x)}$$

15.2 Sine and cosine solution checking

If:

$$y = \sin x$$

then:

$$y' = \cos x$$

$$y'' = -\sin x$$

So:

$$y'' + y = 0$$

If:

$$y = \cos x$$

then:

$$y' = -\sin x$$

$$y'' = -\cos x$$

So:

$$y'' + y = 0$$

15.3 Product with exponential

If:

$$y = xe^{(ax)}$$

then use product rule:

$$y' = e^{(ax)} + axe^{(ax)}$$

$$y' = e^{(ax)}(1 + ax)$$

16. Differentiation Examples to Practice

Example 1

Differentiate:

$$y = 4x^3 - 2x + 7$$

Solution:

$$y' = 12x^2 - 2$$

Example 2

Differentiate:

$$y = x^2 e^x$$

Solution:

Use product rule:

$$y' = x^2 e^x + e^x (2x)$$

$$y' = e^x (x^2 + 2x)$$

Example 3

Differentiate:

$$y = \sin(3x)$$

Solution:

$$y' = 3\cos(3x)$$

Example 4

Differentiate:

$$y = \ln(x^2+1)$$

Solution:

$$y' = 2x/(x^2+1)$$

Example 5

Differentiate implicitly:

$$x^2 + xy + y^2 = 10$$

Solution:

Differentiate term by term:

$$d/dx(x^2) = 2x$$

For xy , use product rule:

$$d/dx(xy) = x \, dy/dx + y$$

For y^2 :

$$d/dx(y^2) = 2y \, dy/dx$$

So:

$$2x + x \, dy/dx + y + 2y \, dy/dx = 0$$

Group dy/dx terms:

$$(x + 2y)dy/dx + 2x + y = 0$$

$$(x + 2y)dy/dx = -2x - y$$

Final:

$$dy/dx = -(2x+y)/(x+2y)$$

17. Integration Examples to Practice

Example 1

Integrate:

$$\int (3x^2 - 4x + 5) dx$$

Solution:

$$= x^3 - 2x^2 + 5x + C$$

Example 2

Integrate:

$$\int (2/x) dx$$

Solution:

$$= 2 \ln|x| + C$$

Example 3

Integrate:

$$\int e^{(5x)} dx$$

Solution:

$$= e^{(5x)}/5 + C$$

Example 4

Integrate:

$$\int \cos(4x) dx$$

Solution:

$$= \sin(4x)/4 + C$$

Example 5

Integrate:

$$\int 2x/(x^2+3) dx$$

Solution:

Let:

$$u = x^2 + 3$$

$$du = 2x \, dx$$

So:

$$\int 2x/(x^2+3) dx = \int du/u$$

$$= \ln|u| + C$$

Final:

$$\ln(x^2+3) + C$$

18. Formula Sheet for Quick Revision

Differentiation

$$d/dx(C) = 0$$

$$d/dx(x^n) = nx^{(n-1)}$$

$$d/dx(e^x) = e^x$$

$$d/dx(e^{(ax)}) = ae^{(ax)}$$

$$d/dx(\ln x) = 1/x$$

$$d/dx(\sin x) = \cos x$$

$$d/dx(\cos x) = -\sin x$$

$$d/dx(\tan x) = \sec^2 x$$

$$d/dx(uv) = uv' + vu'$$

$$d/dx(u/v) = (vu' - uv')/v^2$$

$$d/dx[f(g(x))] = f'(g(x))g'(x)$$

Integration

$$\int x^n dx = x^{(n+1)}/(n+1) + C, \quad n \neq -1$$

$$\int 1/x dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int e^{(ax)} dx = e^{(ax)}/a + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int dx/(1+x^2) = \tan^{-1}x + C$$

$$\int dx/\sqrt{1-x^2} = \sin^{-1}x + C$$

$$\int u dv = uv - \int v du$$

19. Most Important Formulas for Your ODE Chapter

If you are studying first order differential equations, remember these first.

19.1 Separable equations

$$dy/dx = g(x)h(y)$$

Separate:

$$dy/h(y) = g(x)dx$$

Integrate:

$$\int dy/h(y) = \int g(x)dx + C$$

19.2 Linear first order equation

$$dy/dx + P(x)y = Q(x)$$

$$I.F. = e^{(\int P dx)}$$

$$y(I.F.) = \int Q(I.F.)dx + C$$

19.3 Homogeneous equation

If:

$$dy/dx = f(y/x)$$

use:

$$y = vx$$

Then:

$$dy/dx = v + x \, dv/dx$$

19.4 Exact equation

$$Mdx + Ndy = 0$$

Exact if:

$$M_y = N_x$$

Solution:

$$\psi(x,y) = C$$

where:

$$\psi_x = M$$

$$\psi_y = N$$

20. What to Memorize First

If you are a beginner, memorize these first:

Differentiation priority list

1. $\frac{d}{dx}(x^n) = nx^{(n-1)}$
2. $\frac{d}{dx}(e^x) = e^x$
3. $\frac{d}{dx}(\ln x) = 1/x$
4. $\frac{d}{dx}(\sin x) = \cos x$
5. $\frac{d}{dx}(\cos x) = -\sin x$
6. Product rule
7. Chain rule
8. Implicit differentiation

Integration priority list

1. $\int x^n dx = x^{(n+1)}/(n+1) + C$
 2. $\int 1/x dx = \ln|x| + C$
 3. $\int e^x dx = e^x + C$
 4. $\int \sin x dx = -\cos x + C$
 5. $\int \cos x dx = \sin x + C$
 6. $\int f'(x)/f(x) dx = \ln|f(x)| + C$
 7. Substitution method
 8. Integration by parts
-

21. Final Advice

For differential equations, you do not need to know every advanced calculus formula at the beginning.

You must be strong in:

- basic derivatives
- basic integrals
- separating variables
- logarithm rules
- exponential rules
- chain rule
- product rule

If you master these, ODE solving becomes much easier.

End of Formula Master Notes

Prepared as a clean PDF version of the Markdown formula notes.